

In modern scientific terms, the observable universe began about 13.8 billion years ago in an event commonly called the Big Bang. According to general relativity and astronomical observations, that moment marked the sudden emergence of space, time, energy, and matter. From an initial state that was exceedingly hot and dense, the cosmos has expanded and cooled ever since. Edwin Hubble's discovery that distant galaxies have redshifted light – implying they are moving away from us – tells us that space itself is expanding. This expansion, when run backwards in time, means the universe must once have been much smaller and hotter. In fact, the discovery of the cosmic microwave background in 1965 (a faint uniform glow of microwave photons) provided key evidence: it is the cooled remnant of the primeval hot plasma. Likewise, calculations of Big Bang nucleosynthesis (the formation of light elements) predict about 75% hydrogen and 25% helium by mass in the early universe, which matches what we observe in ancient gas clouds today. These observations – the expansion of the universe, the cosmic background radiation, and the abundances of light elements – together form the strongest scientific proof that the universe has a hot, dense beginning, consistent with the Big Bang theory.

Scientists study the universe's earliest moments by combining observations with physics and simulations. In particular, the first tiny fraction of a second is crucial. The inflationary model posits that around 10^{-36} seconds (a billionth of a billionth of a billionth of a second) after the Big Bang, the universe underwent an extremely rapid, exponential expansion. This idea was proposed to explain why the universe appears so uniform on large scales (the horizon problem) and why space is nearly geometrically flat (the flatness problem). Inflation would have stretched any initial inhomogeneities to cosmic size and also magnified quantum fluctuations, seeding the galaxies and large-scale structures we see today. After inflation, the universe settled into a hot dense plasma of particles that gradually cooled. By about 380,000 years after the Big Bang, atoms formed and the universe became transparent, releasing the cosmic microwave background photons we observe. We cannot directly see earlier than this last-scattering surface, but physicists infer what happened by using particle physics and gravity. For example, modern particle accelerators like CERN's Large Hadron Collider recreate tiny fireballs of energy that mimic the state of the universe a millionth of a second after its birth. These experiments let scientists study how quarks and gluons (the building blocks of nuclei) behaved in the early universe. Computer simulations (such as those modeling galaxy formation) also start from initial conditions set by the early universe and predict how it would evolve. All of these methods help us probe as far back in time as possible with our current technology.

However, physics has limits. If one naively extrapolates the Big Bang back to time zero using Einstein's equations, one finds a singularity: a point of infinite density and temperature. In practice this signals that our theories break down at the very beginning. Most physicists agree that a quantum theory of gravity is needed to truly describe the first instants of the universe. In such a theory, concepts like space and time might behave very differently, and it may be meaningless to talk about "before" the Big Bang. At present, the Planck time ($\sim 10^{-43}$ seconds) is the earliest epoch we can meaningfully describe; earlier than that, spacetime may have been so extreme that our usual notions of time and space lost their usual meaning. In other words, the Big Bang is often viewed as the origin of time itself, so asking what came "before" can be tricky or nonsensical under our normal understanding. That said, some theoretical models propose scenarios where something did exist prior to the hot Big Bang phase (see below).

Mathematics and theoretical physics offer further insight into what might underlie the origin. The standard cosmological model is based on the Friedmann–Lemaître–Robertson–Walker (FLRW) solution of Einstein’s equations. These equations describe a homogeneous, isotropic expanding universe and indeed predict an origin. They show that, under reasonable assumptions of matter and energy content, the scale of the universe shrinks to zero at a finite past time – the classical “Big Bang singularity.” Many physicists think this is a sign that the theory has pushed too far and that quantum effects must intervene. For instance, the Hartle–Hawking “no-boundary proposal” uses a mathematical trick (imaginary time) to avoid a sharp beginning. In that picture, the universe is finite but unbounded, analogous to how the surface of a sphere has no edge: the past becomes like a smooth cap rather than a point. String theory and M-theory, which propose extra spatial dimensions and new entities called “branes,” also offer alternative mathematical scenarios. In one popular idea (the ekpyrotic model), our universe arose from the collision of two three-dimensional branes in a higher-dimensional space. In another version of string cosmology, different energy configurations of strings and fields could transition into a growing universe. These approaches are mathematically complex, but they illustrate that purely from a mathematical standpoint there are ways to conceive an origin without a singularity. No single theory is confirmed, however: without experimental evidence, these remain speculative.

Quantum mechanics has a critical role in any origin story. Quantum field theory tells us that even “empty” space (the vacuum) is not truly empty: it seethes with fluctuations and has a nonzero energy density. Some cosmologists have taken this idea further, suggesting that a quantum fluctuation could have given birth to the universe. For example, the notion of a “false vacuum” (a metastable state with high energy) decaying into a lower-energy state can drive inflation. In this view, our observable universe would be a region that popped into existence as that decay occurred. Lawrence Krauss famously argued in *A Universe from Nothing* that quantum fields can create a universe without requiring any pre-existing “thing.” Critics note, however, that even a vacuum with laws of physics is not philosophically “nothing.” Still, quantum mechanics offers one pathway by which the universe might arise “from nothing” in a technical sense – that is, from a state with no classical space but with physical laws. Experimental clues in quantum cosmology are indirect. We know quantum processes were important in the very early universe because they generated the tiny fluctuations that inflation magnified into galaxies. Observing those fluctuations (for example in the detailed patterns of the CMB) is one way scientists see quantum effects at the birth of structure.

Physicists also look for other cosmic relics from the early universe. Besides the electromagnetic background (CMB), there should be a cosmic neutrino background: a sea of neutrinos left over from about one second after the Big Bang, when neutrinos stopped interacting frequently with other particles. These neutrinos would have cooled to just a few kelvin in temperature. We have not yet detected this neutrino background because neutrinos interact so weakly, but in principle it exists and holds information about that first second. There is also expected to be a background of gravitational waves (ripples in spacetime) from the early universe – particularly from inflation. Detecting primordial gravitational waves would be a major breakthrough. Experiments like BICEP and the Planck satellite have searched for the faint polarization signature they would imprint on the CMB. In 2014, BICEP2 announced a detection of such a signal, but it was later shown to be caused by galactic dust, not inflation. Researchers continue to refine these measurements. If primordial gravitational waves are ever found, that would provide direct evidence about processes in the first tiny fraction of a second.

In these ways – looking at CMB patterns, simulating high-energy physics, and searching for faint backgrounds – scientists squeeze out as much knowledge as they can about the origin of the universe.

Within this scientific picture, dark matter and dark energy – the two mysterious dominant components of the cosmos today – also enter the story. Dark matter is an invisible form of mass that holds galaxies together by gravity. It began to clump very early, acting as a gravitational scaffold for normal matter. However, in the first seconds after the Big Bang, normal radiation and ordinary matter dominated the energy density, so dark matter did not drive the initial expansion (though it was present as part of the matter content). Dark energy, which is responsible for the accelerated expansion of the universe today, played essentially no role in the original hot Big Bang. Observations show dark energy became important only after the universe was already billions of years old. Thus dark energy did not influence the origin; it only affects the universe's later evolution (and its ultimate fate). Both dark matter and dark energy are among the great open questions: we do not yet know what they are made of. Whatever their nature, though, they do not contradict the idea of a Big Bang beginning.

How do scientists measure the age and size of the universe? By combining data from many sources. The expansion rate (the Hubble constant) and the contents of the universe allow astronomers to estimate a timeline. The best measurements (for example from NASA's WMAP and ESA's Planck space missions) give about 13.8 billion years. Observations of the oldest star clusters and white dwarfs yield similar ages. Thus the universe has a finite age by all accounts – roughly 14 billion years. Whether space itself is infinite is another question. Measurements of the universe's geometry show it is very close to flat (Euclidean), which is what inflation predicts. A flat universe could be infinite in extent or just very large; current data cannot rule out either. Importantly, a finite age combined with an observed expanding, cooling universe is strong evidence for a beginning. The only mainstream alternative – the "steady state" model of mid-20th-century cosmology – posited no beginning and continual creation of matter. This idea was largely abandoned after the discovery of the cosmic microwave background and the growth of galaxies with time, both of which fit a Big Bang scenario instead.

So, what does all this science say about the Big Bang? It portrays a transformation rather than a creation ex nihilo in the everyday sense. The universe began in a very different state – a hot plasma with radiation, particles, and energy – and then evolved into a cooler, structured cosmos of stars and galaxies. In that sense, the Big Bang was like a massive explosive expansion of an initial condition. If time began with it, this was truly the "start" of cosmic history as we know it. Physics does not answer why that initial condition existed, only that given certain physical laws it evolved in a particular way.

The question of ultimate cause or purpose is something science is not equipped to answer on its own. Many people want to know "Who or what created the universe?" Science describes how the universe changes and what the initial state was like, but it does not say **why** it exists at all. Philosophers and theologians fill this gap in different ways. Some philosophical arguments posit that if the universe began, it must have a cause outside itself. For example, the Kalam cosmological argument (derived from medieval Islamic philosophy) asserts: "Everything that begins to exist has a cause. The universe began to exist. Therefore, the universe has a cause." Proponents often identify this cause as a creator

God. Critics respond that quantum mechanics shows not all events need a classical cause, or that the premise might not apply to the universe. Other arguments invoke the notion of a “necessary” being or principle: something that exists by necessity and is the source of all that exists. These ideas, however, involve concepts that lie outside empirical science. Science itself simply notes that if time began at the Big Bang, causality as we know it may not apply at that boundary.

Related to this is the question of why there is something rather than nothing, or why we exist at all. Science observes that the constants of nature seem finely “tuned” for complexity and life: small changes in some constants would prevent stars or chemistry from forming. This has been noted by physicists such as Martin Rees and others. Some see this fine-tuning as a hint of design: they argue an intelligent creator set the values. Others invoke the anthropic principle: we should not be surprised that the universe allows life, because if it didn’t, we wouldn’t be here to notice. In the context of a possible multiverse (many universes with different constants), the anthropic principle is often used to explain fine-tuning without design. Mathematically, if infinitely many universes exist, ours might just be one of the few hospitable ones. But the multiverse idea is itself highly speculative and untestable so far. In any case, mainstream science remains neutral: it acknowledges the fine-tuning problem but usually regards it as an open question or an area for further theoretical insight.

What do religions say about the origin? Most religions have creation stories. In Judeo-Christian tradition, the Book of Genesis begins: “In the beginning God created the heavens and the earth.” Many believers interpret this as God speaking the universe into existence over six “days” (which some take literally as 24-hour days, others as long ages). Islam has a similar concept, with Allah creating the universe. Hindu cosmology, by contrast, envisions time as cyclical: vast eons (called kalpas) of creation, preservation, and dissolution repeat endlessly. A well-known metaphor is the cosmic egg or the god Brahma dreaming the universe into being and later dissolving back. Buddhism generally does not posit a single creation moment; time may be infinite and cyclic with no first cause. Native American myths vary by tribe: for example, some say the Earth was formed on the back of a turtle or by a Great Spirit molding the land. Ancient Greek mythology starts with a primordial Chaos, from which Gaia (Earth) and Uranus (Sky) emerged, later giving birth to the pantheon of gods. Norse myth speaks of the Ginnungagap, the great void between icy Niflheim and fiery Muspelheim; from this void arose the giant Ymir, and when the gods later slayed him, Ymir’s body became the earth. Egyptian mythology often tells of a primeval ocean (Nun) from which the sun god (Ra or Atum) emerged and created other gods and the world. Dreamtime stories of Australian Aboriginal culture describe ancestral beings shaping the world in a timeless realm that underlies reality. Even older African myths include creation by divine beings or actions (for example, some West African stories involve the Yoruba deities crafting the world). In Babylonian myth (the “Enuma Elish”), the storm god Marduk kills the chaos serpent Tiamat and forms the sky and earth from her body. In Zoroastrian tradition, the god Ahura Mazda creates the world in a cosmic struggle with the destructive spirit Angra Mainyu. All these accounts share a common feature: they involve conscious agents or forces creating the universe out of chaos or nothing. They are richly symbolic and meaningful within their cultures, but they do not resemble scientific explanations.

Today, many people hold views that mix science with spirituality. Atheists or agnostics typically accept the Big Bang and subsequent evolution of the cosmos as a natural process without any divine

involvement. Deists believe a creator set up the universe but does not interfere; they often accept the Big Bang as the method of creation. Pantheists or panentheists see God and the universe as identical or interwoven, so the cosmos itself is a manifestation of the divine; in such views, the origin is both natural and sacred. Some religious individuals endorse “theistic evolution” – the idea that God used the Big Bang and evolution as tools, reconciling sacred texts with scientific findings. Others (such as Young Earth creationists) insist the universe is only a few thousand years old and reject most of modern cosmology; they often have their own calculations based on scripture. Polls (for example in the United States) find that views vary widely: a majority of people accept that the universe had a beginning, but many are split on whether God played a role. In more secular societies (Europe, East Asia), acceptance of the Big Bang is higher; in more religious societies (some parts of the US, Africa, Middle East), many still prefer creationist accounts. Overall, public belief is a patchwork: some trust science alone, some trust religion alone, and many try to combine both.

Throughout the last century, the scientific story of cosmic origin has itself evolved. Early in the 20th century, many physicists assumed the universe was eternal and unchanging (a “steady-state” cosmos). Edwin Hubble’s discovery of expansion (1920s) challenged that. In the 1940s, physicists like Gamow, Alpher, and Herman realized a hot early universe would naturally explain observed element abundances. In the 1950s, Sir Fred Hoyle coined the term “Big Bang” (intended derisively) for this idea. For a time, an alternative “steady-state” theory (Hoyle, Bondi, Gold) competed, which held the universe has no beginning and constantly creates new matter. The decisive evidence against steady-state was the cosmic microwave background (1965) and the evolving distribution of galaxies. Since then the Big Bang model (specifically the modern Λ CDM model with inflation) has been overwhelmingly supported. Remarkably, even what we mean by “Big Bang” has shifted: it used to mean a singular explosion, but now it usually refers to the early hot, dense phase; inflation and quantum gravity push any singular beginning to a “before.”

Modern cosmologists continue to seek evidence about the first instants. One goal is detecting primordial gravitational waves, ripples in spacetime created by inflation. Instruments like the BICEP and QUBIC telescopes, as well as future space-based observatories, aim to measure the polarization of the cosmic microwave background for the telltale “B-mode” patterns. If found, these would be strong proof of inflation and give clues about physics at energies far beyond terrestrial experiments. Similarly, cosmologists search for any sign of ancient cosmic strings or topological defects, hypothetical leftovers from early phase transitions in the universe’s structure; these would be evidence of high-energy particle physics in the first seconds. Large-scale computer simulations (such as the Millennium or Illustris projects) reconstruct cosmic evolution from just after the Big Bang to the present. They take initial conditions (tiny density fluctuations) and evolve them under gravity and other forces; the result remarkably matches the web-like distribution of galaxies observed today. This gives us confidence that our understanding of early conditions is on the right track. But there are limits: we cannot see back earlier than the CMB except indirectly (via neutrinos or gravitational waves), and any theoretical statement about “before” requires assumptions.

Philosophy can step in where empirical science currently stops. Some thinkers argue that if the universe has a beginning, perhaps time itself emerged at that point and asking about “before” is nonsensical (similar to asking what is north of the North Pole). Others debate whether the universe

requires a cause at all. The arrow of time is another puzzle: the second law of thermodynamics says entropy (disorder) increases, yet the Big Bang was a state of incredibly low entropy (very uniform and simple). Why did the universe start in such an ordered state? Some propose that a deeper physical principle or “initial condition” must explain it. Others suggest that our very existence as observers may bias our perspective. Questions like “Why is there something rather than nothing?” or “Does the concept of cause apply at the origin?” are as much philosophical as scientific.

It is worth noting that some modern speculative ideas push the boundaries of science. The “multiverse hypothesis” suggests our universe is one of countless others, each perhaps with different physical constants. In this view, the Big Bang was just a local event in a much larger (possibly eternal) reality. Eternal inflation is an example: while one region stops inflating and forms a universe like ours, other regions keep inflating indefinitely. If true, the question of a single origin becomes moot, since there might be no first moment overall. The simulation hypothesis is another out-there idea: it imagines that an advanced civilization could have simulated our universe on a powerful computer, so what we call the Big Bang might actually be the booting up of that simulation. This is a philosophical possibility but currently has no way to be tested. Similarly, idealist or mystical philosophies sometimes claim that consciousness or mind underlies reality, and that physical laws emerged from a cosmic mind. These views are not part of mainstream science, but they illustrate the range of human thinking about origins.

Let us also revisit some of the specific questions in light of these perspectives. When we ask “Who or what created the universe?”, scientists would answer in terms of natural processes: given the laws of physics, the universe could have arisen from certain conditions without needing a personal creator. Religious believers often answer “God” (or gods) by faith. Science does not rule out a deity, but it does not require one to explain the Big Bang. If pressed, a physicist might say “We do not know of any physical entity that ‘created’ the universe – it might have begun as a quantum fluctuation or as a boundary condition.” On the other hand, a theologian might say “God created it ex nihilo.” Both statements describe different kinds of explanation.

Similarly, “Why are we here?” in a scientific sense is answered by recognizing that we are here because the universe happened to produce stars, planets, and chemistry in the right way. We exist simply because life emerged given the conditions. Many religious traditions would say we are here for a purpose bestowed by a creator. Science remains agnostic on “why” beyond “it’s a consequence of laws,” whereas philosophy may try to assign meaning. Both perspectives are explicit about their answers, even if they differ.

We have covered the Big Bang in simple terms already. Another point: some people wonder if the universe is infinite in age or had a beginning. Current science strongly indicates a finite age (13.8 billion years) for our observable patch, with no evidence of a prior infinitely ancient state. Even if space is infinite, time seems to have a starting point (in standard cosmology). Religious and philosophical answers vary: some believe in an eternal universe (like certain Hindu or Buddhist views) while others insist on a first moment.

Other concepts: The nature of time at the origin is subtle. In relativity, time is just another dimension, so the phrase “beginning of time” is tricky. If time itself came with the Big Bang, there was literally no “when” before it. Some theories (like the block universe or eternalism) regard time as an unchanging four-dimensional whole, so in that view the origin is simply one part of an eternal spacetime. That philosophical stance implies the question of a beginning may not even arise in the usual sense. By contrast, most physical discussions treat the Big Bang as a beginning of time.

Could our universe arise from “nothing”? Physicists have debated this phrase. If “nothing” means no space, no time, no matter – then no, because in physics we always start with a mathematical framework (fields, quantum laws). But if “nothing” means a quantum vacuum with no particles, then yes some theories allow a universe to tunnel into existence. The Hartle-Hawking no-boundary proposal says the universe had no edge, and some loop quantum gravity models suggest a “bounce” from a previous contracting universe. In fact, loop quantum cosmology predicts a bounce that replaces the singularity: a high-density state on one side leading to expansion on the other, so the universe might have oscillated. String-inspired models like the ekpyrotic scenario also avoid a singularity by involving extra dimensions. These are mathematical and conceptual ways of conceiving an origin without a true “beginning from nothing” as a one-off event.

Regarding philosophical questions like the Kalam argument, it’s worth noting that modern physics suggests the usual logic of cause and effect may break down at the quantum or Planck scale. So applying classical cause-and-effect to the universe’s birth may be invalid. That is one reason some scientists think the question of “who created God?” is meaningless – if God is defined as necessary and uncaused, then God is taken outside the normal chain of cause and effect. Analogously, the universe itself might be self-contained in a way that physics doesn’t yet understand. Some have argued that asking “who caused the universe” is like asking “who caused the big bang” which is a little misguided since the Big Bang is not an object but a state.

On the fine-tuning problem: many scientists simply note that the constants of nature allow observers; they either regard it as a curiosity or invoke ideas like the multiverse. Others see it as evidence of design. Again, both perspectives are explicit: the multiverse hypothesis is one way to explain fine-tuning naturally, whereas design implies a deliberate creator. Neither can currently be tested, but both are clear answers that people give to the question of why the universe seems so conducive to life.

In mathematics, infinity plays a role too. For example, some cosmological models have an infinite past or future. But infinities often lead to paradoxes (like Hilbert’s hotel). If time were truly infinite in the past, the universe might be infinitely old and have no beginning, which conflicts with the observed finite age. Current evidence favors a finite age. Infinity more commonly arises in the possible infinite size or energy of the universe. Mathematics handles infinity well, but physically an infinite past is problematic.

Are there any paradoxes of causality at the origin? Possibly. If time starts, then cause and effect have no “before.” In quantum mechanics, some interpretations allow events that are not determined by prior causes (truly random outcomes). So the idea that everything must have a cause can break down at the quantum level. This lends some support to ideas like Krauss’s, that a universe can pop into existence without a cause in the everyday sense. But these ideas remain highly theoretical.

Let us now consider some alternative or fringe ideas. The steady-state theory (now abandoned) was one alternative scientific model: it held the universe looked the same at all times, with continual creation of matter. By the late 1960s, it fell out of favor because it couldn’t explain the CMB or the evolving universe. Plasma cosmology was another rival idea (championed by Hannes Alfvén) that tried to use electromagnetic forces in plasma to form structures, but it never gained mainstream acceptance. Today, most credible models extend the Big Bang by adding things like inflation or cycles, rather than rejecting a beginning altogether. For example, in cyclic models like Penrose’s Conformal Cyclic Cosmology (CCC), each remote future of an expanding universe becomes the Big Bang of a new universe. According to CCC, our Big Bang was actually just the far future of a previous “aeon” seen in a different light. Penrose has even claimed to see faint imprints of past aeons in the CMB (though this claim is debated). Other cyclic ideas include “big bounce” models from loop quantum gravity and the ekpyrotic brane collisions. All these aim to solve problems like the low-entropy beginning by positing an eternal sequence of universes. If any of these are correct, the universe never truly “started” in an ultimate sense – it just goes through endless rebirths. So whether the universe has a single beginning or is part of an infinite cycle is still unanswered by science.

What do surveys say about public understanding? Generally, scientific literacy surveys reveal that many people know of the Big Bang, but misconceptions persist. Some still think of it as an explosion in space (rather than an expansion of space). Polls of belief often find that in many countries a majority accepts the Big Bang (usually with varying belief about God’s involvement). For example, a Pew survey found that many Americans say they believe in the Big Bang, but a significant fraction add “with God’s help.” Meanwhile, some people genuinely hold creationist views regardless of evidence. In any case, public belief on origins is varied: in scientific communities, the Big Bang is almost universally accepted; in religious communities it depends on interpretations; the general public covers the spectrum from strict creationism to atheistic cosmology. Clearly, what “people generally believe” depends heavily on culture, education, and religion.

Finally, we should note what questions remain open and what future discoveries might contribute. As mentioned, the particle physics experiments and astrophysical observations continue to refine the age and contents of the universe. Detection of primordial gravitational waves would be a watershed event confirming inflation. Discovering the cosmic neutrino background would open a new window to the first seconds. Measuring the geometry and topology of the universe better could tell us if space is truly infinite or wraps around in some way. On the theoretical side, finding a consistent quantum gravity theory could tell us whether the Big Bang was a true beginning or a transition. And answering the nature of dark matter and dark energy might even have unexpected implications for cosmology. Until then, science provides a clear and detailed story of how the universe expanded from a hot state to the present: it tells us *when* (13.8 billion years ago), *what* (expanding space, primordial plasma, formation of atoms), and *how* (gravity, quantum fields, nuclear physics) the cosmos evolved.

Science does **not** inherently provide a “why” or a “who” – those questions are answered differently in philosophy and religion, where people speak of purpose or deity.

In summary, our understanding of the universe’s origin spans many perspectives. Modern cosmology gives a detailed chronological picture from the first moments after the Big Bang onward, supported by observations of expansion, background radiation, and elemental abundances. Mathematical physics offers models that sometimes remove the notion of a strict beginning (via cyclic or quantum models). Philosophy and theology offer reasoning about cause, necessity, and meaning, often pointing to a creator or deeper principle. Creation myths from ancient cultures provide symbolic accounts where modern science does not tread. Each perspective is explicit in its own way: science explicitly describes a Big Bang without invoking purpose; religion explicitly invokes divine creation; philosophy explicitly debates causes or denies meaningful questions. No single viewpoint fully closes the matter. The ultimate origin of the universe – the “before” of the Big Bang or the reason for existence – remains one of the greatest open questions, blending physics, mathematics, philosophy, and spirituality. Advances in observation and theory continue to refine the scientific story, even as humanity’s quest for deeper meaning continues. In the end, the universe’s beginning is both a triumph of science and a profound mystery that touches on our place in the cosmos.